

Quiz1 2026

CADD (BIO563)

Total Marks:20

Date: 14.09.2026

Time: 1 Hour

Section A(1 mark each)

1. The target-validation hierarchy taught in class runs, in order of increasing confidence, as:
 - A. Hunch < Literature precedent < Anecdotal findings < Cell model
 - B. Hunch < Anecdotal findings < Literature precedent < Cell model < Animal model < Pharmacology in animal model < Pharmacology in human disease
 - C. Anecdotal findings < Hunch < Cell model < Literature precedent
 - D. Cell model < Animal model < Anecdotal findings < Hunch
2. Which of the following is the MOST plausible cause of a false positive during hit validation of a biochemical binding assay?
 - A. Genuine, reproducible, target-specific affinity of the compound
 - B. Compound aggregation, optical/spectral interference, or non-specific binding to the assay components
 - C. Use of a recombinant protein expressed in a cell line
 - D. An insufficient number of patients in a Phase III trial
3. A binding pocket is known to change shape substantially upon ligand binding (an induced-fit effect). Which docking strategy is most appropriate to model this?
 - A. Rigid docking, because it is computationally cheap
 - B. Flexible docking, because it can accommodate conformational change in the binding site
 - C. Homology modeling alone, with no docking at all
 - D. Docking is not applicable to flexible pockets
4. A researcher notices that a handful of patients taking a blood-pressure drug for an unrelated condition happen to report improved mood. With no further data collected, this observation sits at which level of the target-validation hierarchy?
 - A. Pharmacology in human disease
 - B. Anecdotal findings
 - C. Validated animal model
 - D. Literature precedent
5. When evaluating a potential drug target, which level of validation provides the strongest evidence and the lowest risk for further drug development?
 - A. Literature precedent
 - B. Pharmacology in human disease
 - C. Pharmacology in the animal model of the disease
 - D. Anecdotal findings

6. A promising compound has been identified, but the team needs to understand what happens to it after entering the body. Which of the following specialists should the team consult?
- A. Automation specialist
 - B. Medicinal chemist
 - C. Molecular biologist
 - D. Pharmacokinetics specialist
7. According to the provided industrial framework, what is the typical duration required to advance a project from the initial idea to the point of 'candidate selection'?
- A. 3-5 years
 - B. 9-16 years
 - C. 20+ years
 - D. 1-2 years
8. Which level of protein structure is stabilized mainly by hydrogen bonds between backbone atoms, giving rise to motifs such as the alpha helix and beta sheet?
- A. Primary structure
 - B. Secondary structure
 - C. Tertiary structure
 - D. Quaternary structure
9. In a standard PDB ATOM record line, which of the following fields is NOT normally present?
- A. Residue name and chain identifier
 - B. x, y, z Cartesian coordinates
 - C. Occupancy and B-factor (temperature factor)
 - D. The compound's IC50 or binding affinity value
10. Which of the following leads advance to lead optimization
- A. Molecules with acceptable potency, physical and ADME properties
 - B. Molecules that have affinity for target molecules
 - C. Molecules that show strong biological activity but poor ADME properties
 - D. Molecules that are highly selective for the target but have unacceptable toxicity

Section B(2 mark each)

1. 'Drugs are overpriced.' Using the cost-of-development and attrition figures discussed in class, critically evaluate this statement. In your answer, explain how the pricing logic differs between a patented drug and its generic equivalent once the patent expires.
- Ans: Uses cost/attrition figures correctly (e.g., \$0.5–2 billion average cost; only ~1 in 30,000 compounds yields a satisfactory ROI) to argue the 'overpriced' claim is more nuanced than it appears.[1 mark]

Correctly explains that patented pricing recoups R&D, trial, and attrition costs across the whole pipeline (including failed candidates), whereas generics enter post-patent-expiry, reusing the proven formulation without bearing those costs — hence their much lower price. Full marks for balanced reasoning acknowledging both sides (e.g., market exclusivity vs. access-to-medicine concerns). [1 mark]

2. Define target validation. Using the hierarchy taught in class, explain why 'anecdotal findings' are considered weaker evidence than 'cell model' data.

Ans: Correct definition: target validation is a form of risk assessment that establishes confidence a modulated target will produce a beneficial disease effect; better validation lowers project risk.

[1 mark]

Correct reasoning: anecdotal findings are uncontrolled, unreplicated human observations, while cell-model data come from a controlled, reproducible experimental system placing anecdote below literature precedent and cell models in rigour. [1 mark]

3. Using the SSRI (antidepressant) example discussed in class, explain how the hypothesis 'preventing reuptake of a neurotransmitter from the synaptic cleft will produce an antidepressant effect' was translated into a druggable target, and describe how that target would then be validated.

Ans: Correct mechanism: blocking the serotonin reuptake transporter keeps serotonin longer in the synaptic cleft, increasing/prolonging postsynaptic receptor stimulation and producing an antidepressant effect. [1 mark]

Correctly links this to target identification (the reuptake transporter becomes the defined target) and validation (collecting public-domain evidence, developing in vitro/in vivo systems, and expert consultation to confirm the hypothesis before hit identification). [1 mark]

4. A student opens a .pdb file in Notepad and sees a line such as: 'ATOM 1 N MET A 1 20.154 29.699 5.276 1.00 49.05 N'. Identify what the residue name, chain identifier, x/y/z coordinates, and B-factor represent in this line. What would it mean if this line instead began with 'HETATM'?

Ans: Correctly identifies: residue name = MET (methionine), chain identifier = A, coordinates = 20.154, 29.699, 5.276 (x, y, z in Angstroms), B-factor = 49.05 (the second-to-last number; temperature/displacement factor). [1 mark]

Correctly explains that HETATM would mean this atom belongs to a non-standard group (a ligand, cofactor, ion, or water molecule) rather than a standard amino acid of the protein chain. [1 mark]

5. Differentiate between a 'hit' and a 'lead' in the drug discovery process. What additional properties must a hit satisfy to be advanced to lead identification?

Ans: Hit: a compound with activity at a predetermined level against the target, but little else known about it. [1 mark]

Lead: a hit that has additionally been shown to have acceptable selectivity, physicochemical/drug-like properties, and metabolic properties (e.g., half-life), making it suitable for optimization. [1 mark]

